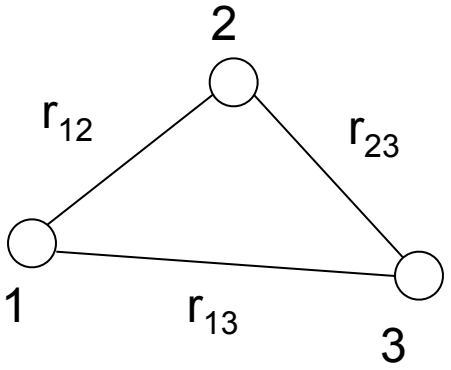
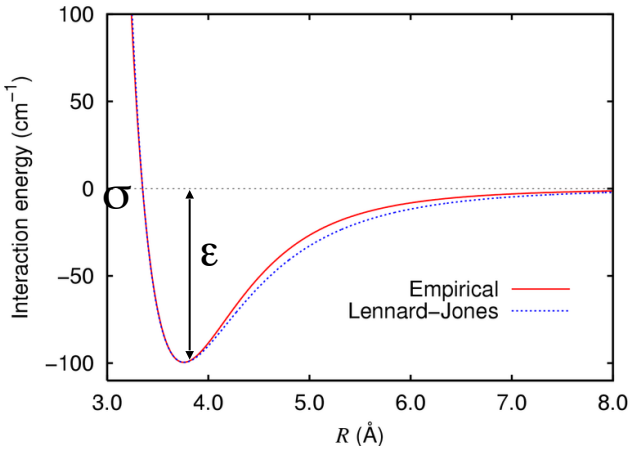
Project: SHAKE algorithm for Ar

- System composed by 200 molecules, with 3 Argon atoms each (m_{Ar}=39.948 g/mol)

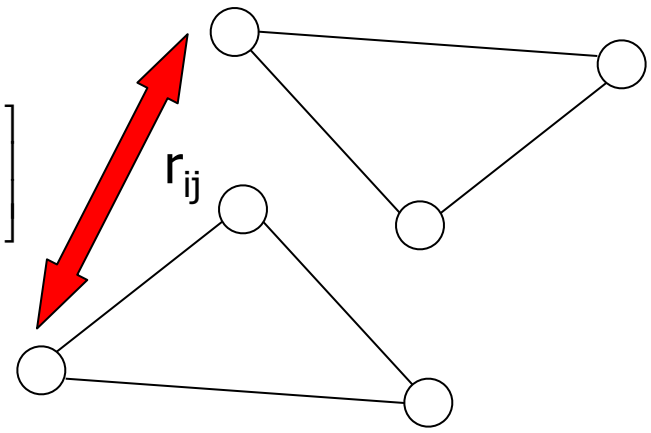


- Atoms inside each molecule do not interact but always keep the geometry of an equilater triangle (SHAKE) with side 3 Å

- Atoms belonging to different molecules interact by a Lennard-Jones
- potential with parameters ε = 119.8 K and σ=3.405 Å



$$V(r_{ij}) = 4\varepsilon\left[\left(\frac{\sigma}{r_{ij}}\right)^{12} - \left(\frac{\sigma}{r_{ij}}\right)^{6}\right]$$



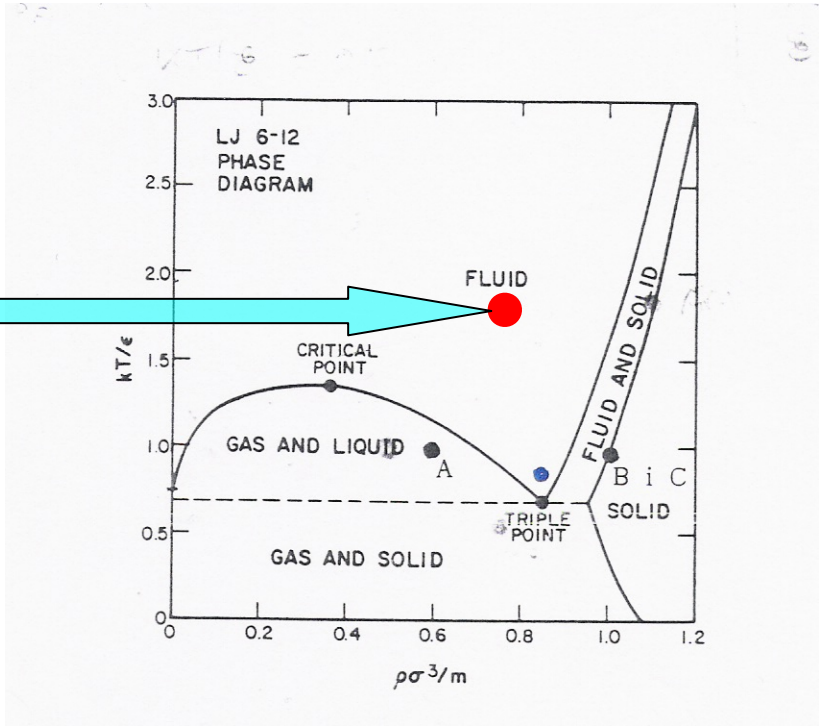
Simulation details

- Thermodynamic conditions:

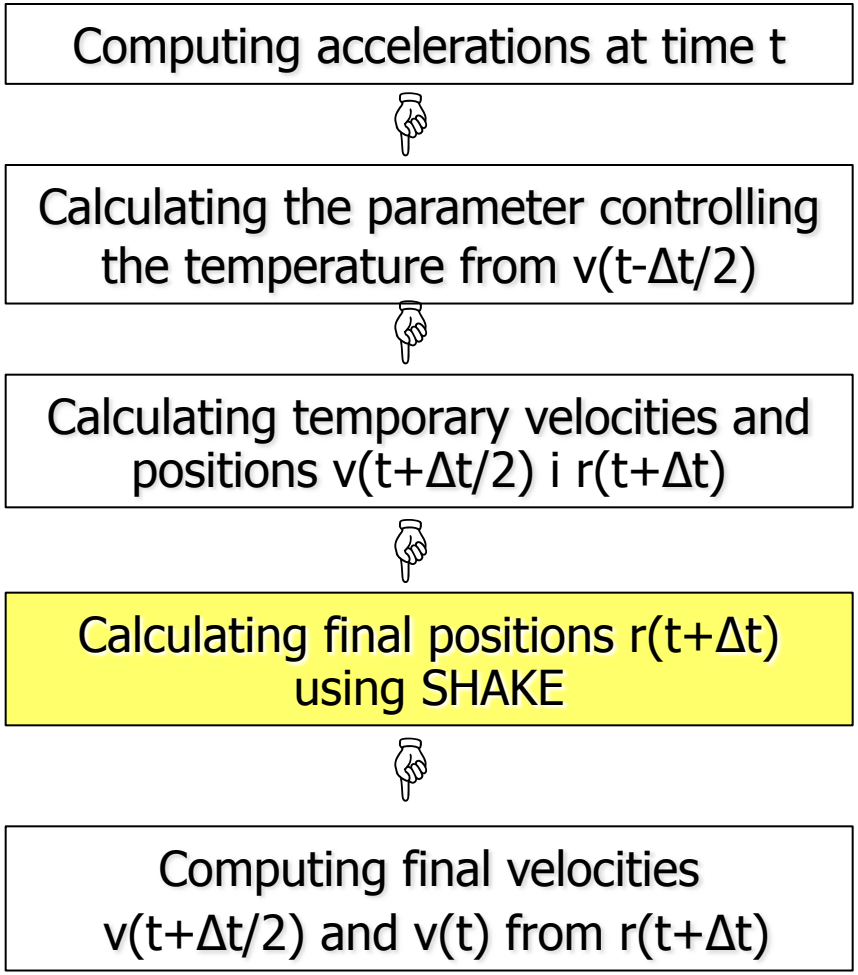
  T=225 K and ρ=0.019 atoms/Å^3

  T^*=1.9 and ρ^*=0.75

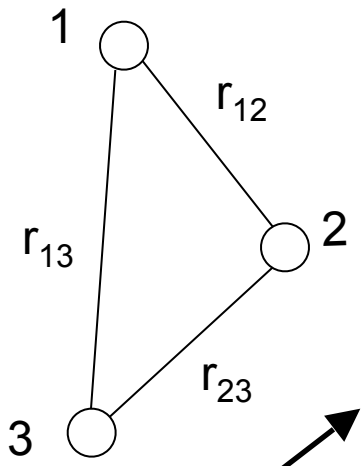
- Integration algorithm: leap-frog Verlet with Δt=0.01 ps and coupled to a thermal bath (Berendsen) with τ_T=0.1 ps



Structure of the code



SHAKE subroutine



Molecule i

Atoms 1,2 and 3 from molecule i

Applying SHAKE to the pair 1,2
r_1 i r_2

Applying SHAKE to the pair 1,3
r_1 i r_3

Applying SHAKE to the pair 2,3
r_2 i r_3

For all pairs:

if $|r_{ij}^2-d^2| >$ tolerance

if $|r_{ij}^2-d^2| \leqslant$ tolerance

Goal: Write a subroutine to keep the triangle structure of the molecule, using SHAKE method

- Create and incorporate the subroutine that keeps the triangle strucutre of each molecule, using SHAKE

- Find within the folder "Exercise_SHAKE":
  - A MD code at constant T without SHAKE
  - Data and dimension files
  - First configuration

- Be aware that the code works internally in reduced units

Questions:

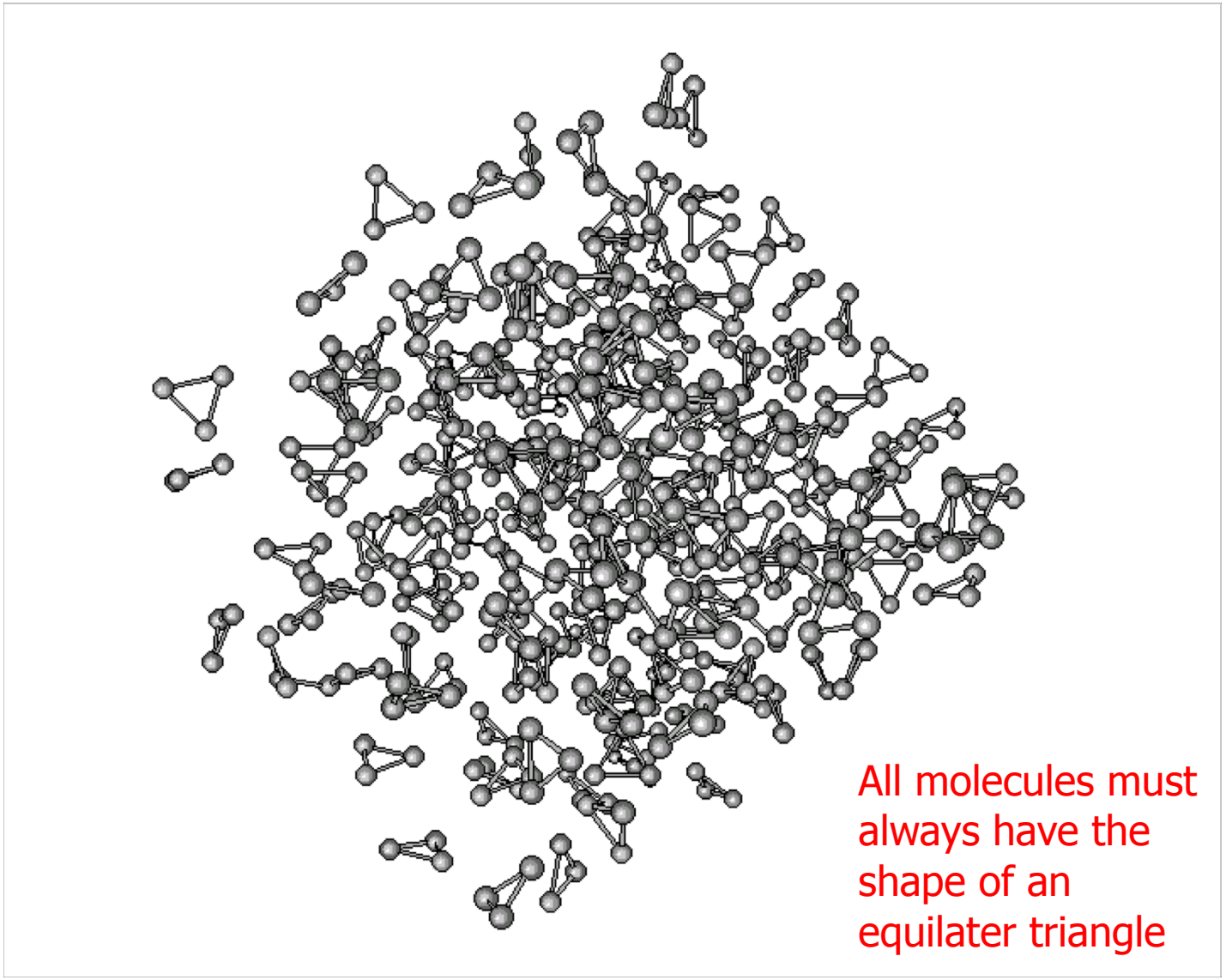
1. Compare energy fluctuation in a system "with SHAKE" vs. "without SHAKE". Which one is the stablest?

2. Derive the Lagrange multiplier of the SHAKE algorithm for a pair of particles, neglecting terms of order $(\Delta t)^4$. Show that it is:

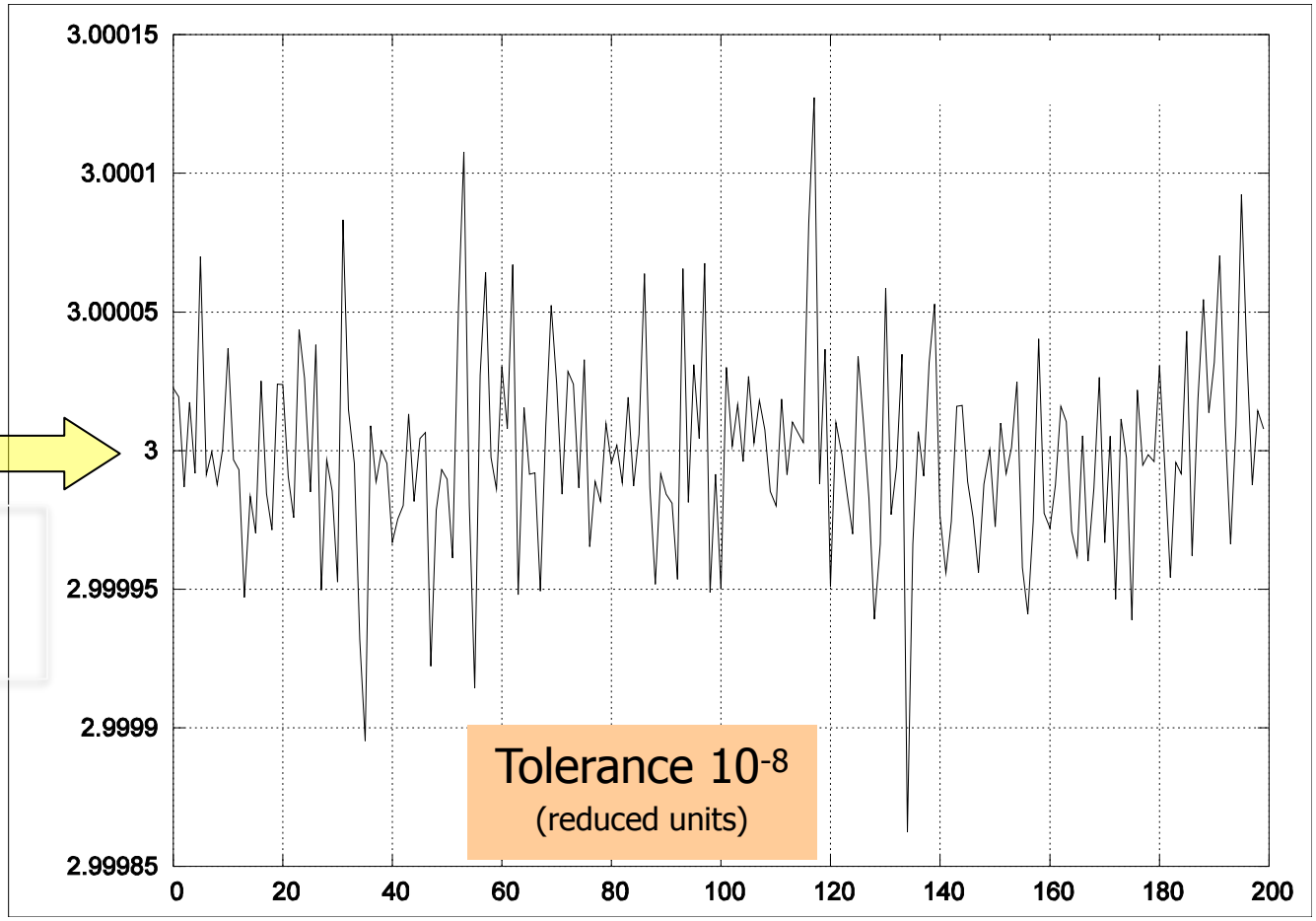
$$\lambda = \frac{r_{p12}^2(t+\Delta t) - d^2}{4(m_1^{-1} + m_2^{-1})\Delta t^2 \mathbf{r_{p12}}(t+\Delta t) \bullet \mathbf{r_{12}}(t)}$$

3. The RATTLE algorithm is quite equivalent to SHAKE, how do they compare in terms of efficiency and accuracy? HINT: Seek the RATTLE algorithm and apply it to your Ar system.

Visualization of the system



Simulation results



3 Å

error
0.01% !!!